LAB REPORT: ANALYZING NETWORK LAYERS IN A SERVER ROOM ENVIRONMENT

COMP1323 Networks and Security
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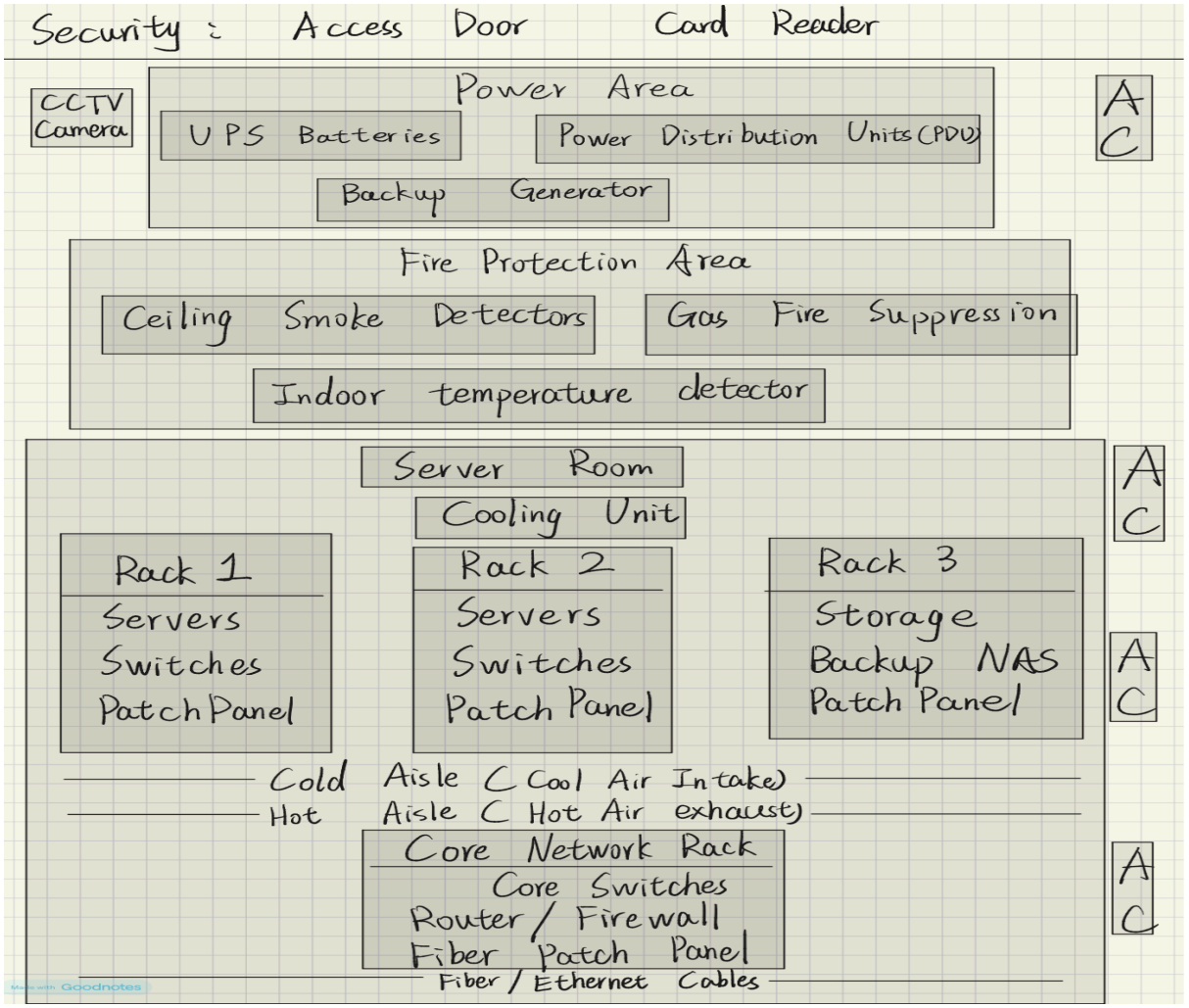
1. Introduction

This lab report combines observations from a guided tour of a real-world server room at the University of Southampton Malaysia with the theoretical framework introduced in Chapter 1 of the COMP1323: Networks and Security textbook. The purpose of this lab exercise is to categorize the most important devices in the server room into the correct layers of the Internet protocol stack (Application, Transport, Network, Link, and Physical).

2. Observation Summary

The server room had server racks in excess of 4 server cabinets, power distribution units, air conditioning units, KVM switches, and network switches. The cooling system was based on airflow. And each server had two power supplies, which were connected to separate PDUs. The network infrastructure was offered by stacked switches with dual uplinks. Besides, the physical security measures included ID card readers and surveillance cameras. The devices were connected using structured cabling and fiber optic cables.



[1]

[1] Diagram of the server room layout

3. Device Classification by Protocol Layer & Performance and security discussion

Below is the mapping of observed devices to the five-layer Internet protocol model:

a) **Physical Layer**
Copper Cables: These are used to connect servers, switches, and PDUs in the rack. They physically carry the electrical signal that corresponds to the bit.
Fiber Optic Cables: These cables are used for high-speed inter-switch and uplink connections to the overall network (e.g., to the core routers or ISP).
Racks and cabinets: These are designed to hold, organize, and protect IT hardware.

b) **Link Layer (Data Link Layer)**
Network Switches (Ethernet Switches): These are layer 2 devices that use MAC addresses to forward frames.
KVM Switch: As a management equipment, the KVM switch uses USB and HDMI over Ethernet connectivity, which is an endpoint for console access.
Network Interface Cards (NICs): These are found in each server and switch and are responsible for encapsulating/decapsulating frames.

The link layer provides reliable delivery between directly connected devices. Redundancy is provided by link aggregation and switch stacking (logical single switch).

c) **Network Layer**
Routers: The default gateway of the server and overall connectivity to the internet are provided by routers that are physically located upstream.

IP Addresses: Each server and switch has its own IPv4 or IPv6 address, which proves that the network layer works. At this layer, decisions about where to send packets (based on IP headers) are made.

The network layer lets hosts on different subnets talk to each other directly. Having two network interfaces on each server shows that multipath routing is being used for redundancy.

d) **Transport Layer**
Servers and Storage Devices: These host applications that use TCP (e.g., web servers, databases) or UDP (e.g., DNS). Even though no transport layer devices are standalone hardware, the functionality resides in the operating systems and applications running on servers.
Redundant Power Supplies and Dual NICs: While physical, their role as fault-tolerant components supports the reliability of the TCP transport protocol, ensuring session continuity despite hardware fault.

The transport layer (TCP/UDP) enables process-to-process communication. Dual NICs allow separate TCP sessions over distinct paths, enhancing availability.

e) **Application Layer**
Servers as Hosts: Web servers (Apache/Nginx), database servers (MySQL), virtualization hosts (VMware), and network management tools all operate at the application layer.
KVM Switch (Management Interface): Provides HTTP/HTTPS-based remote console access, which can be used for server management.
Monitoring Tools: Actively collect and transmit performance data (CPU, temperature, power) to monitoring applications.

These applications provide specific services to end users: email, websites, cloud storage, leveraging lower layers for transport. For instance, if we would like to run and test some shell scripts on Linux, then the remote server provides an opportunity to connect to the University's server at any time and anywhere.

4. Connection to Chapter 1 Concepts

Network Edge: The server room is the network edge. More specifically, the data center network segment where servers (end systems) run applications that can be accessed through the Internet.

Network Core: The routers and core switches that forward packets over the Internet comprise the network core. The server room is connected to the core through high-bandwidth (10-100 Gbps) links.

Redundancy and Layering: Redundancy (dual PDUs, dual NICs, dual ACs) is provided across multiple layers: power (physical), NICs (link), routing (network), and TCP session management (transport).

Encapsulation: Every packet transmitted from a server follows the traditional encapsulation process: Application data → TCP header → IP header → Ethernet header → Bitstream

Security: ID card access for administrators and CCTV cameras support Chapter 1's stress on physical security as part of a defense-in-depth strategy. Network security (firewalls) is logically located upstream but shapes server room design.

5. Conclusion & Personal Reflection

This tour of the server room made the abstract ideas of network layering easier to understand. It's simple to see which layer in the TCP/IP model each device, from the cable to the KVM interface. Redundant parts are not only extra hardware, but also necessary for

the architecture to make sure that services are always available at the physical, link, transport, and application layers. As a result, this hierarchical structure not only makes it clear how data moves through networks, but it also shows why a failure in one layer (like a power outage) can spread unless redundancy design principles are used to stop it.